WILMINGTON UNIVERSITY

COLLEGE OF TECHNOLOGY

SEC 210 – Principles and Practice of Information Security

12/6/2020	Tyler Sanderson

OBJECTIVES: Perform **Lab 3.17B Network Traffic Analysis with Wireshark.** This is found in the **Lab Manual** on pages **271 – 278,** which has good background information. You should read the pages as you perform this lab.

INSTRUCTIONS: Carefully follow the instructions on each page. You will submit numerous screenshots. Search for INSERT or SCREENSHOT in uppercase to find out the exact pages. There are questions to answer. Search for QUESTION in uppercase to find the exact pages.

NOTE: You do not need to reference the Lab Manual. This lab document is self-sufficient.

PREREQUISITES:

- "How to use the Snipping Tool on Windows 10" video at https://www.youtube.com/watch?v=ayi5-7QPNcA
- "How to Take a Screenshot on a Mac" video at https://www.youtube.com/watch?v=h3RkDPfphSQ
- How to access SEC 210 vApp.docx

COMMENTS: I do NOT want a screenshot of your entire VM desktop window when only a portion is needed to demonstrate that you have followed the instructions. For examples, see the screenshots in this document.

Ensure that you complete the entire lab. You will know when you see "THIS IS THE LAST PAGE OF THE LAB".

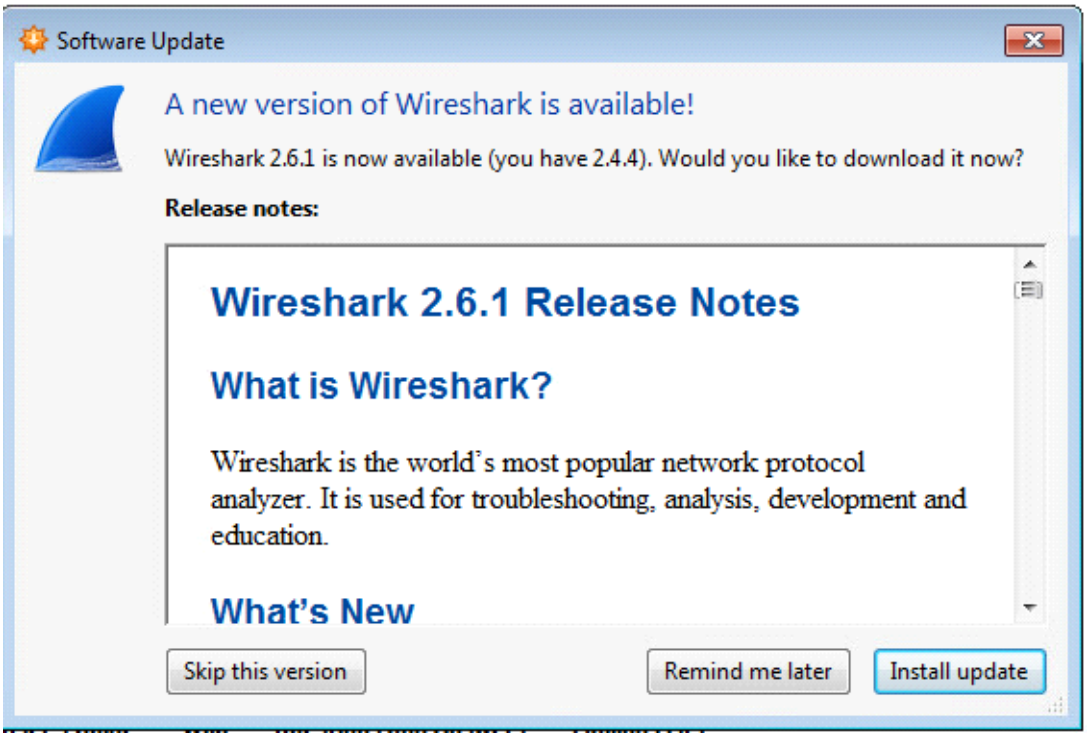
In this lab, you will perform some simple tasks using **msconfig**. To provide more background about this lab and to learn more about other capabilities, watch the video at

https://www.youtube.com/watch?v=jvuiI1Leg6w

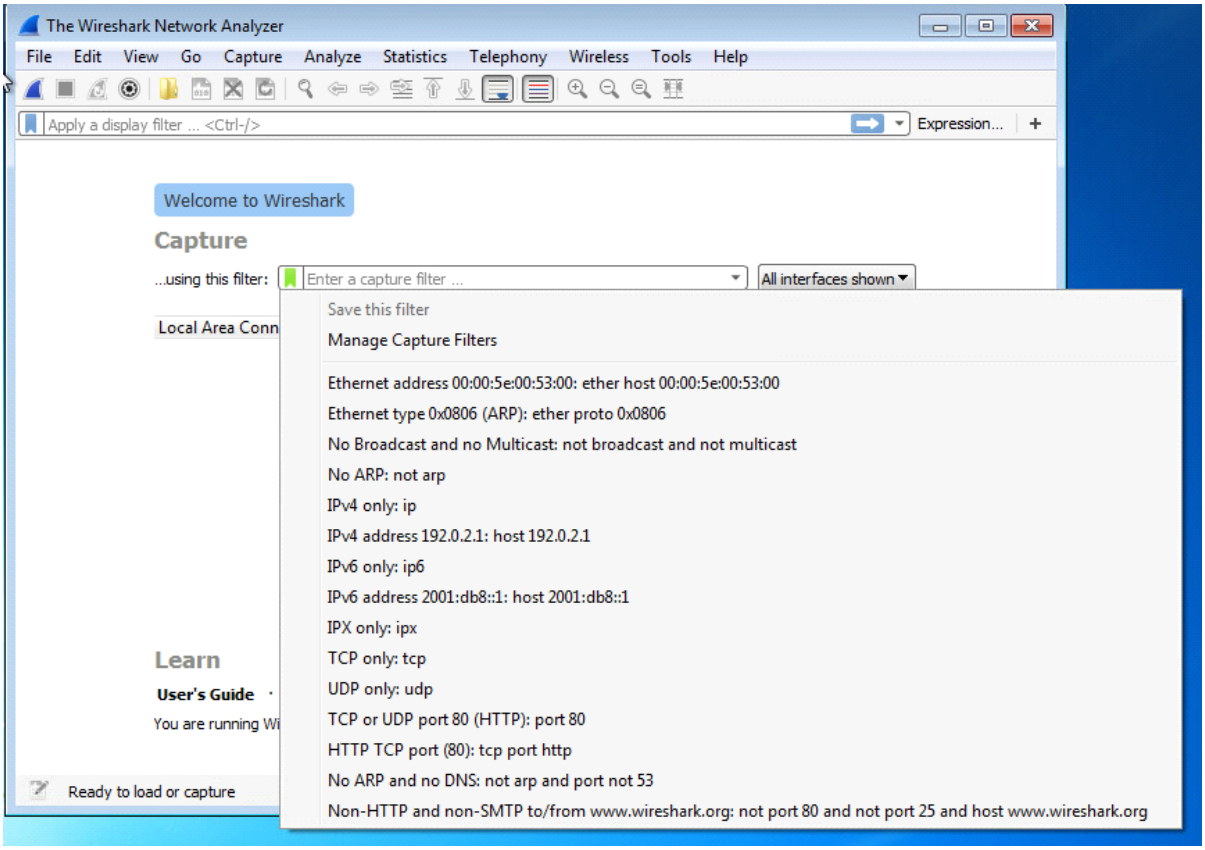
Intro to Wireshark: Basics + Packet Analysis! (16:07)

- Login to the **CNSLabWin7** VM. The password is "**cnslab**"
- You may need to resize the VM to expose the taskbar at the bottom of the window
- NOTE: **CLOSE ALL OPEN BROWSERS BEFORE PROCEEDING**
- Click on the **Wire Shark** icon on the task bar at the bottom of the window

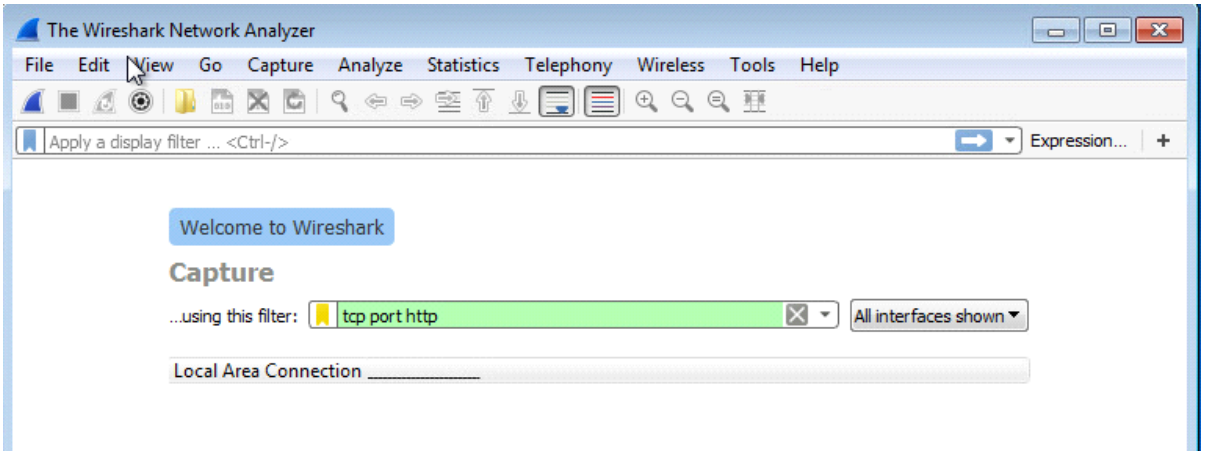
- A Software Update window appears
- Click on **Skip this version**



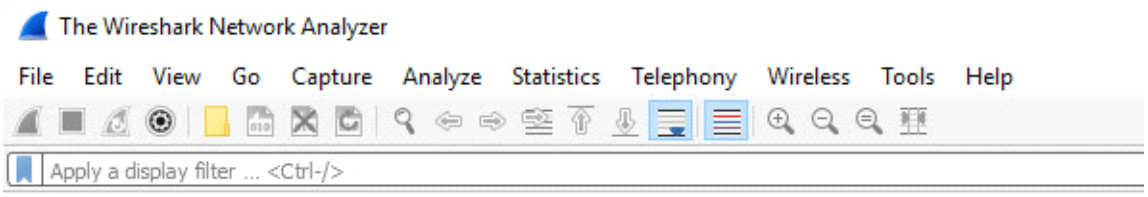
- The following screen appears.
- Click on the **green** icon to the left of the **Capture** field
- A drop-down list is shown
- Select **HTTP TCP port (80) tcp port http**



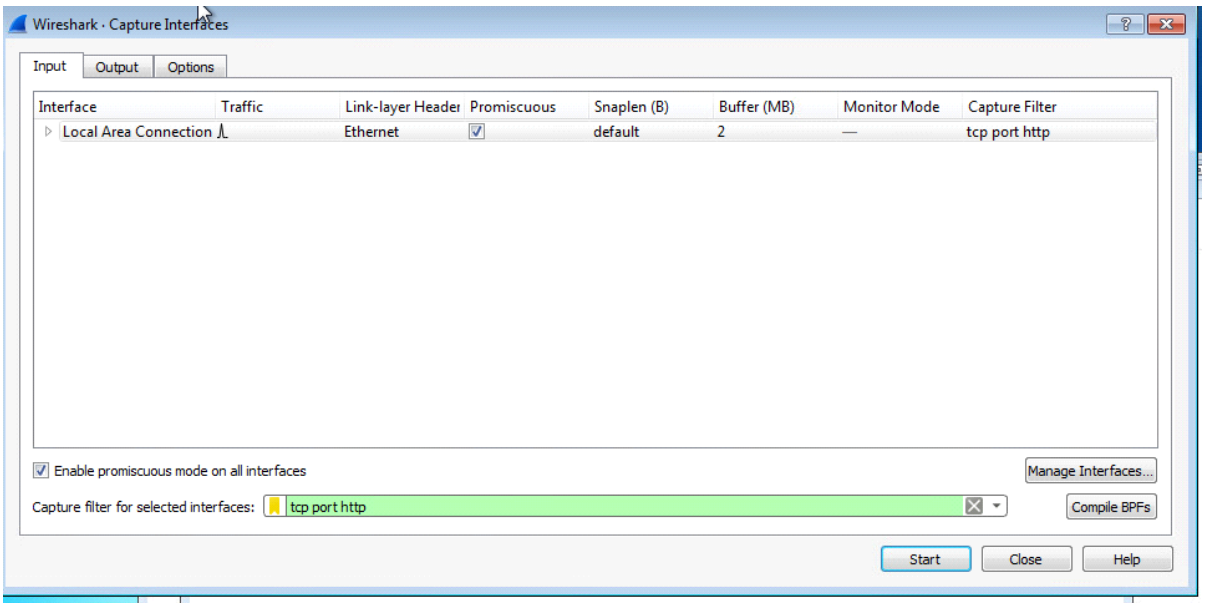
- Now the **Capture** field has a value



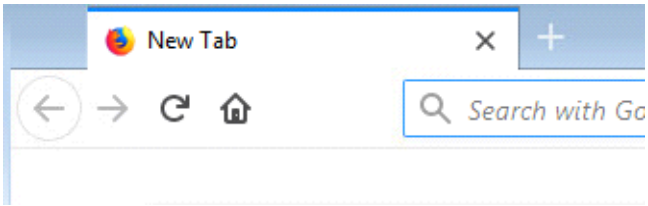
- Click on the **Capture options** icon on the tool bar



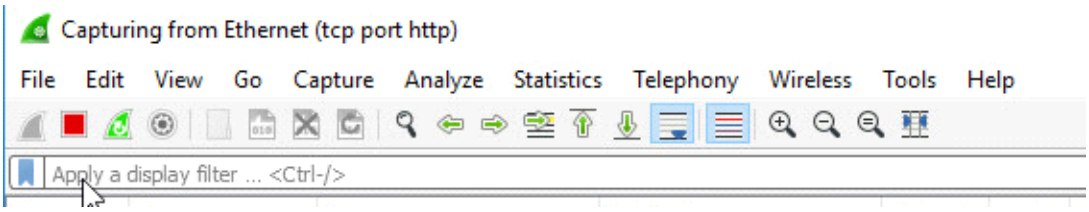
- The **Capture Interfaces** window appears
- Select **Local Area Connection**
- Click on **Start**



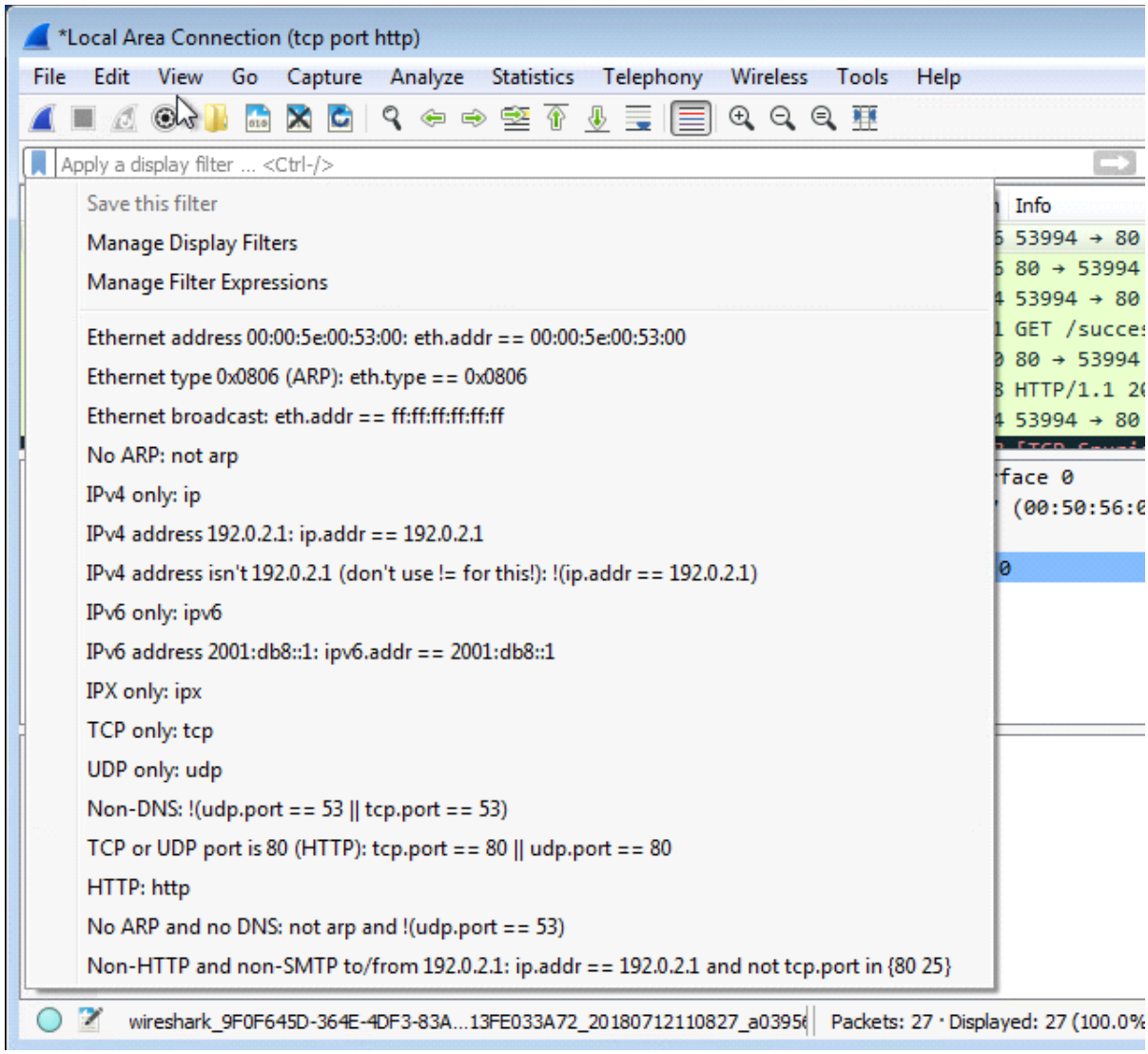
- The **Capturing from Ethernet** window appears
- Open **Firefox** to start some HTTP and TCP network traffic
- Click the **Home** button



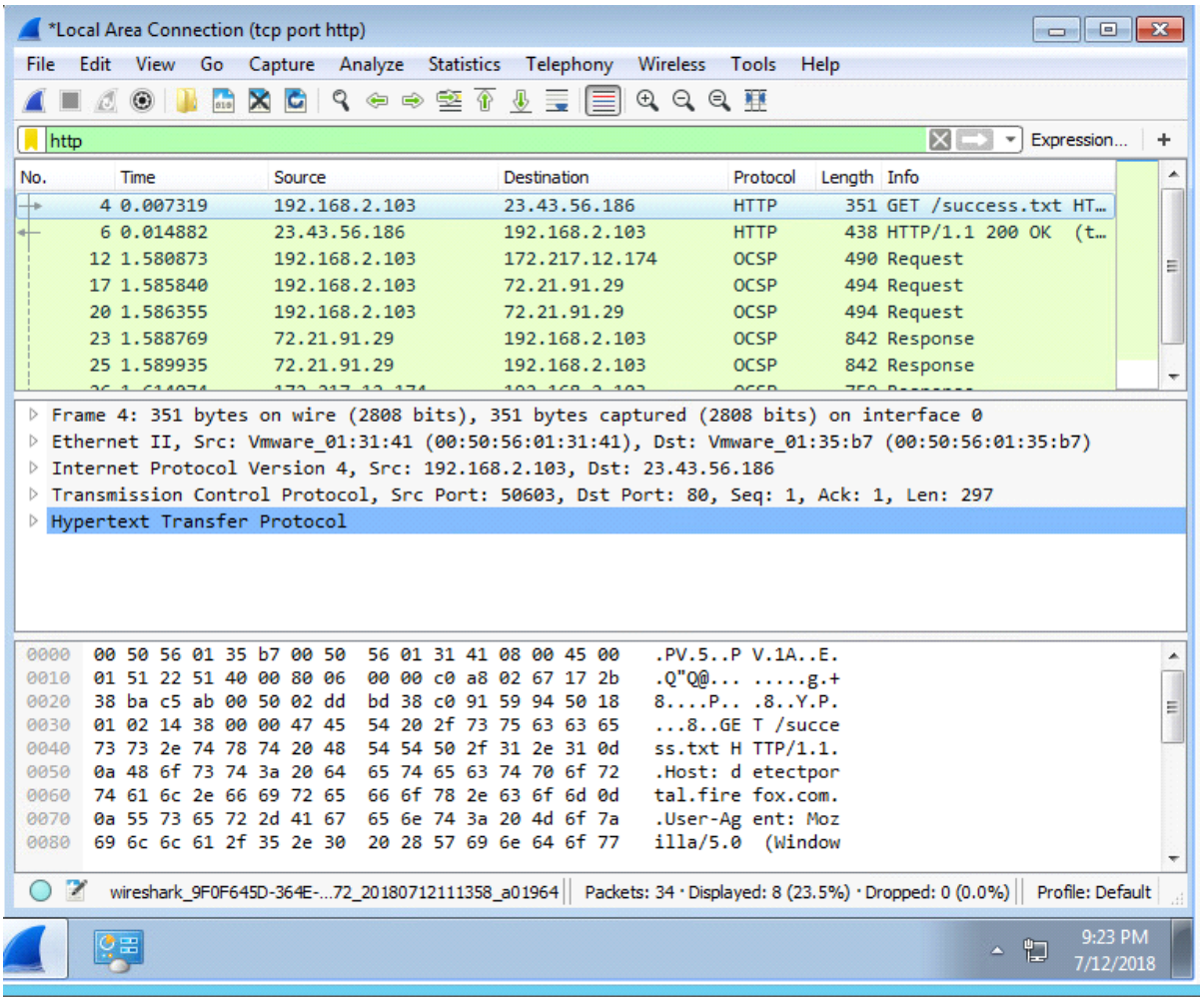
- After at least 50 packets have been captured, click on the **Stop** icon on the toolbar. HINT: Use the vertical scroll bar on the right to scroll down keep track of the number packets



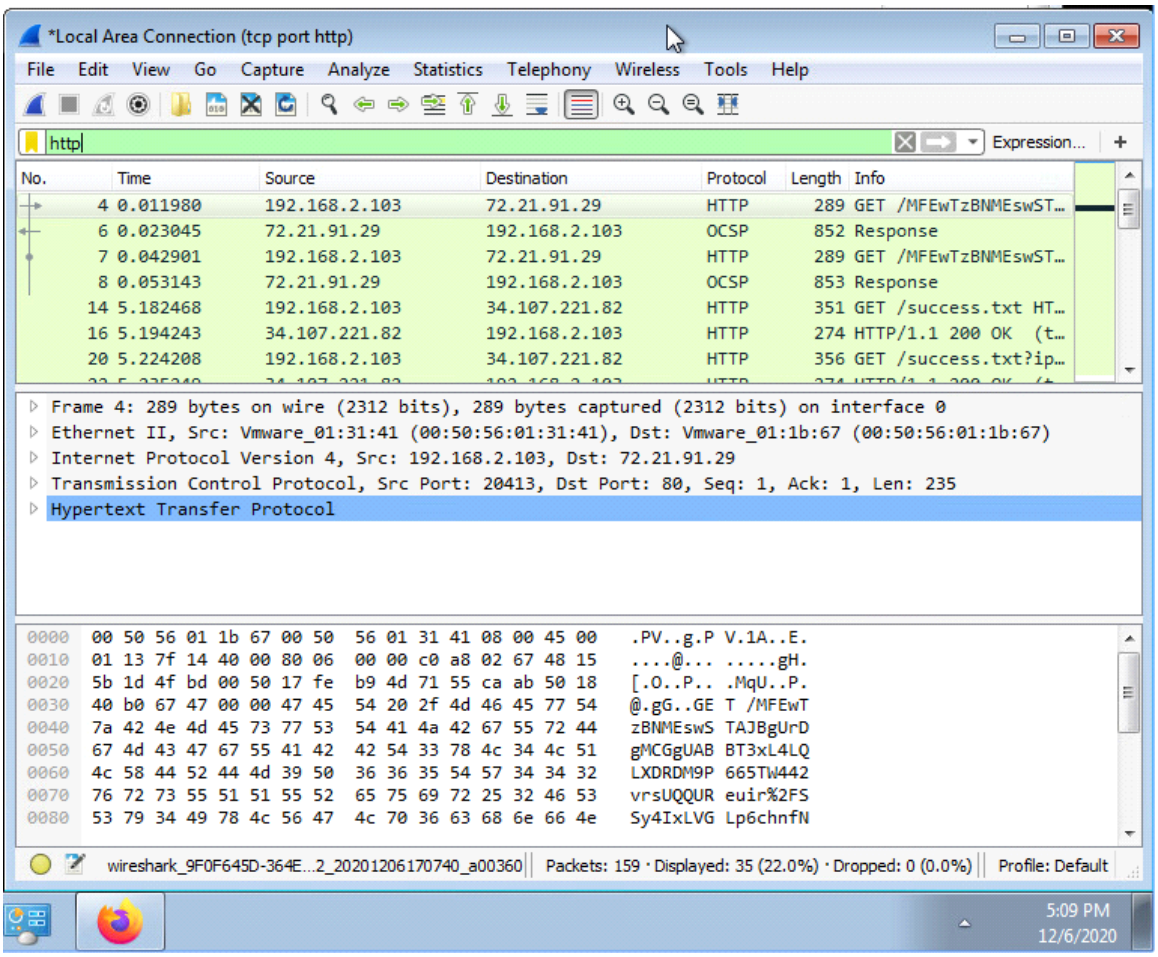
- Click the **display filter** drop-down icon
- Select **HTTP: http**



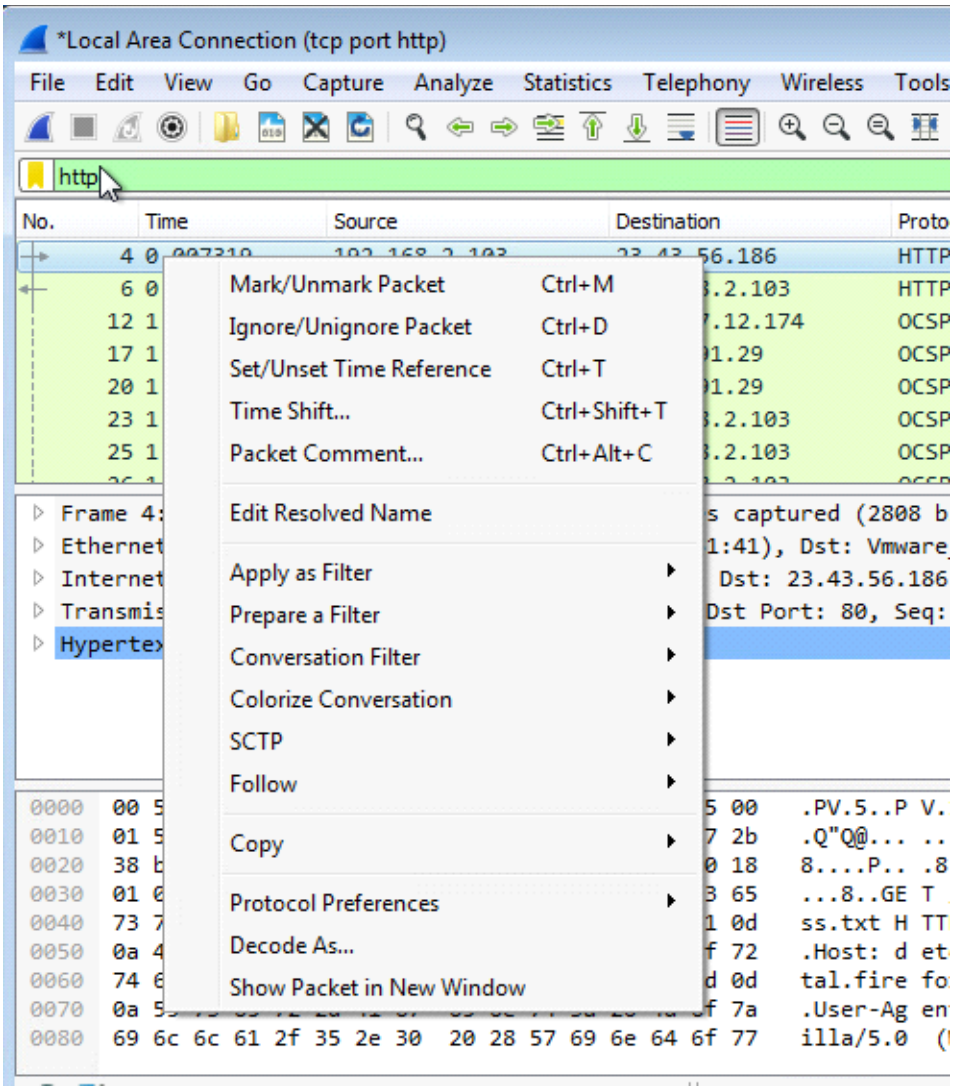
- The **Local Area Connection** window appears
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like



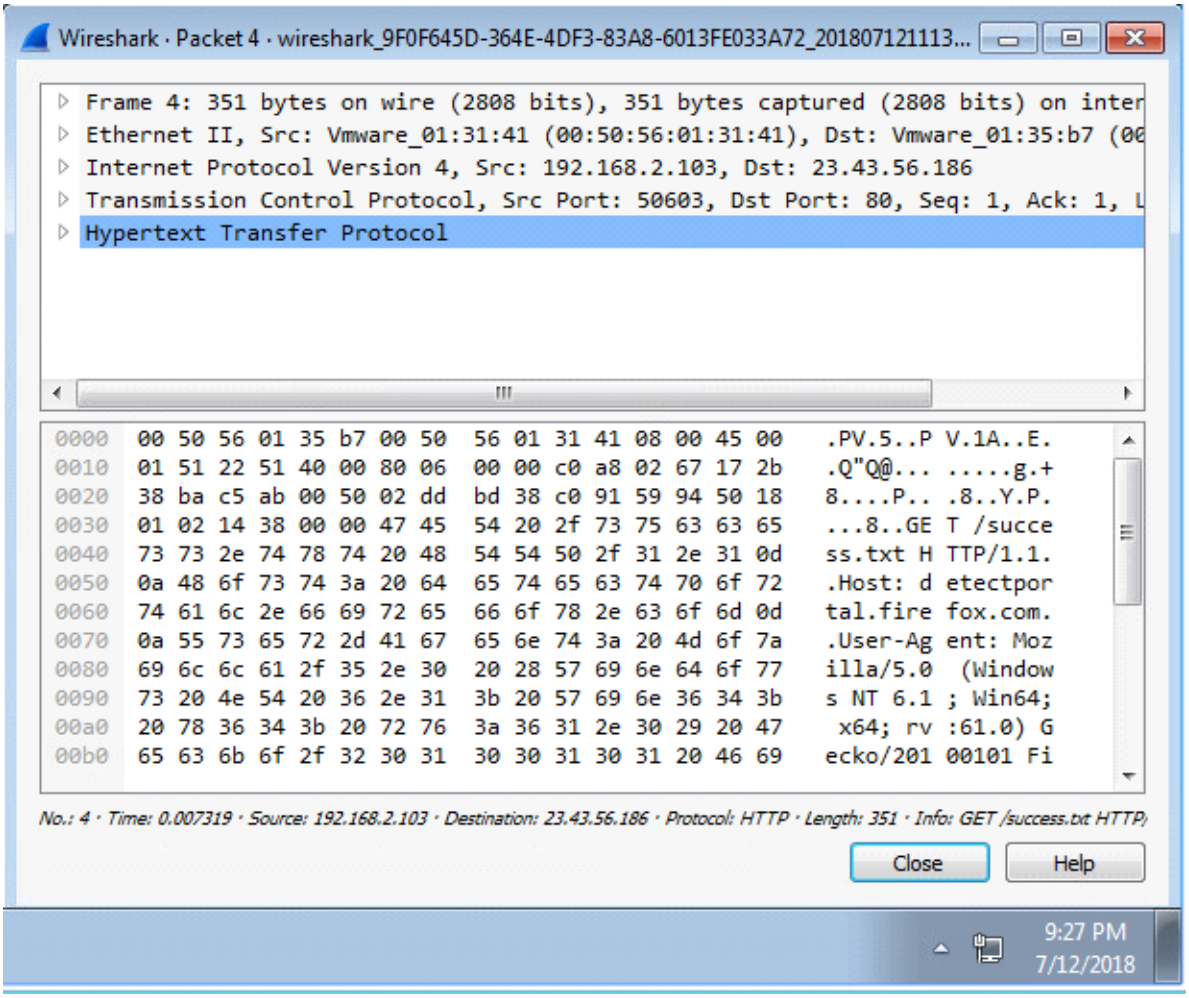
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***15 POINTS***



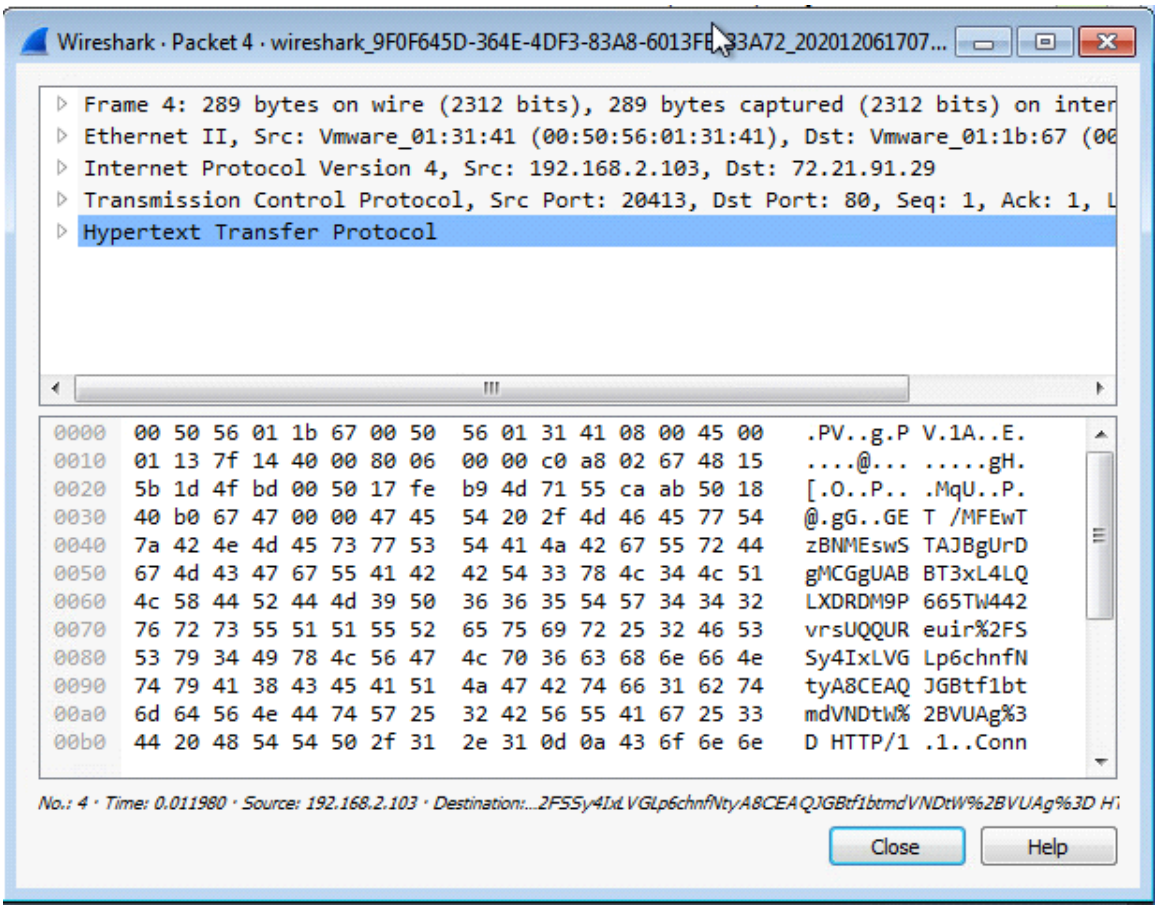
- Select the first entry where **Source** is **192.168.2.103** and **Protocol** is **HTTP**
- Right-click on the entry to show a pop-up menu
- Select **Show Packet in New Window**



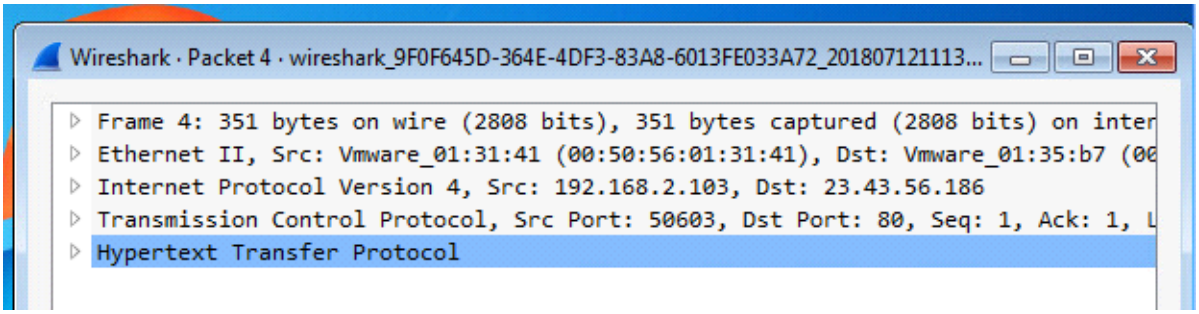
- A **Packet** window appears
- Move the **Packet** window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like



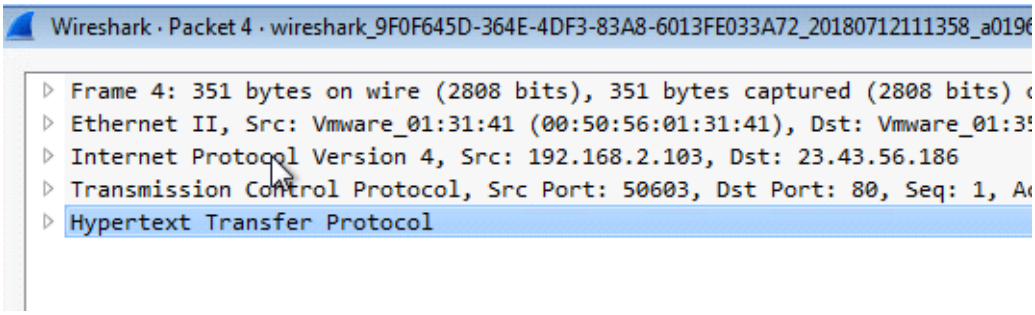
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***15 POINTS***



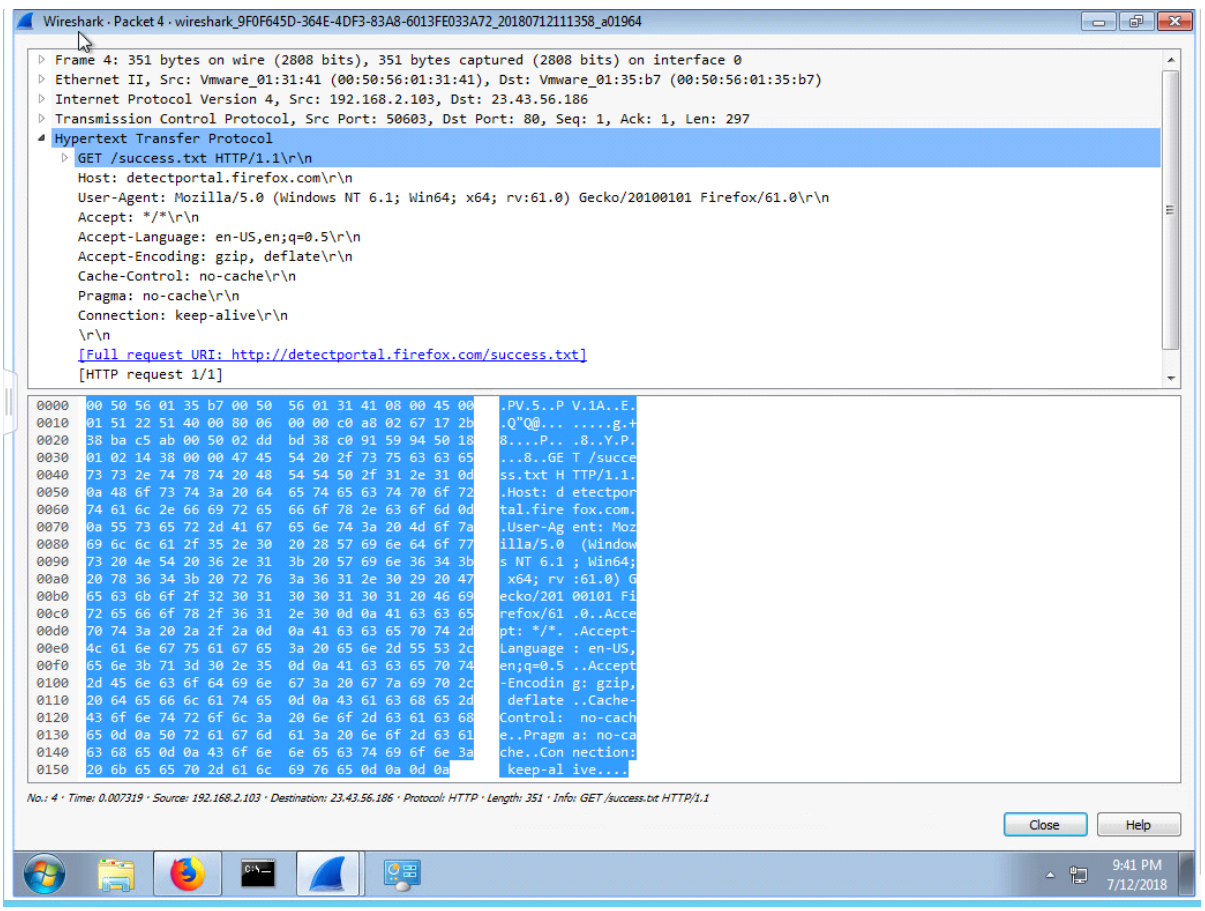
- Maximize the Packet window



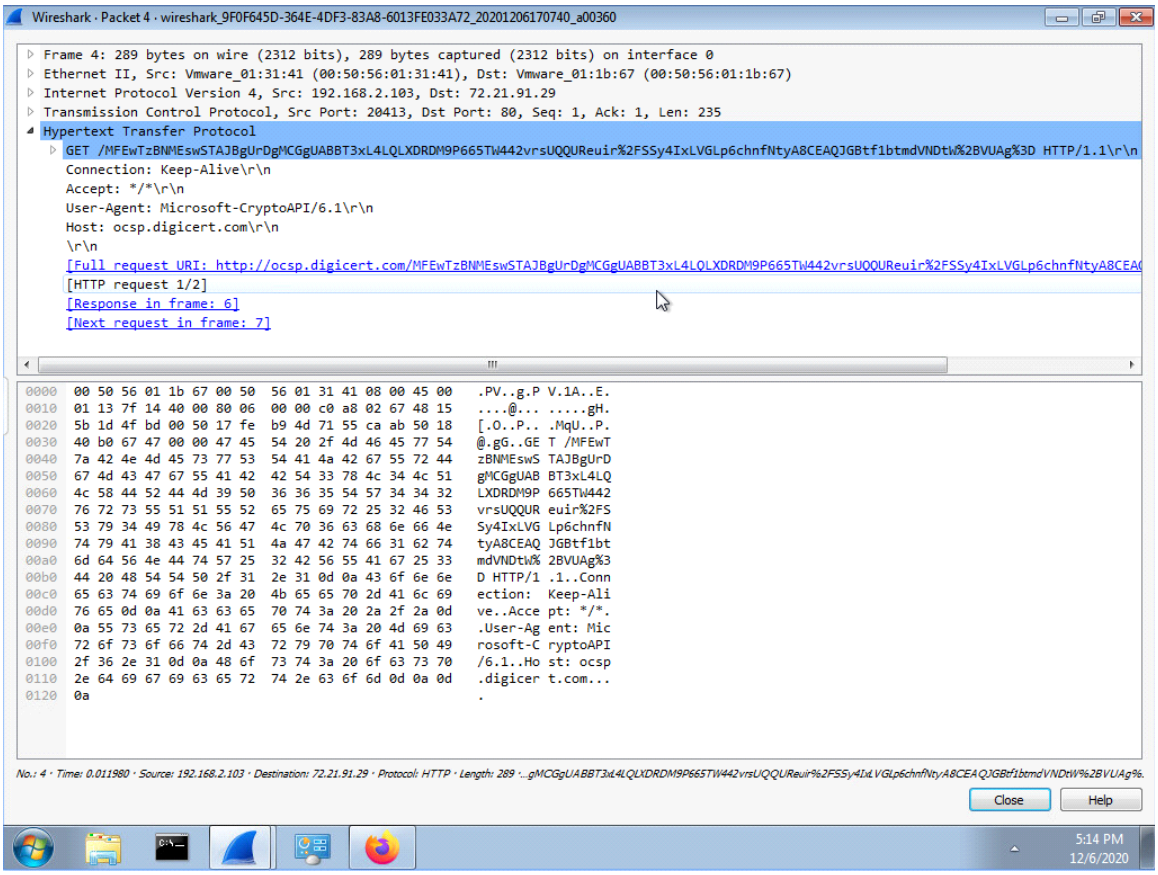
- Click on the arrow to the left of **Hypertext Transfer Protocol** entry



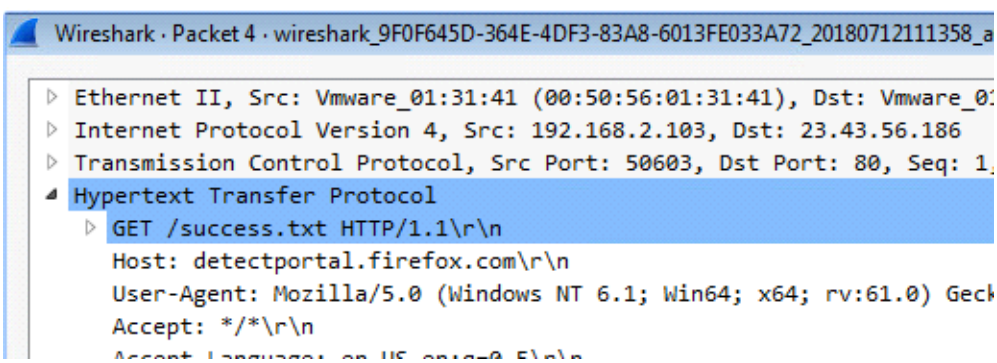
- The entry expands
- Using Snipping Tool, take a screenshot that includes the date/time. It should look like



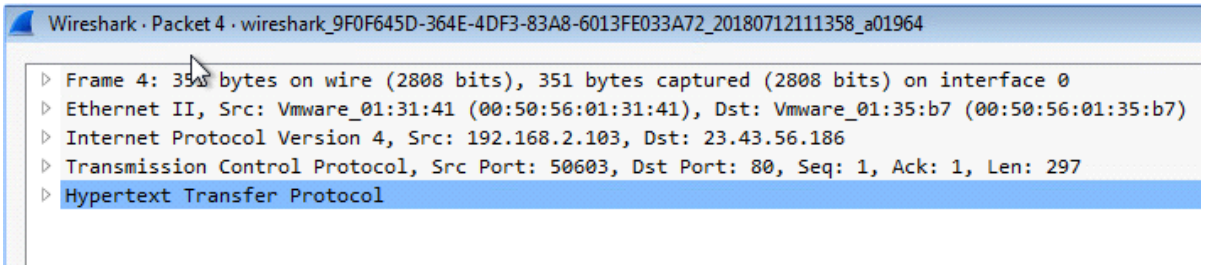
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***<u>15 POINTS</u>***



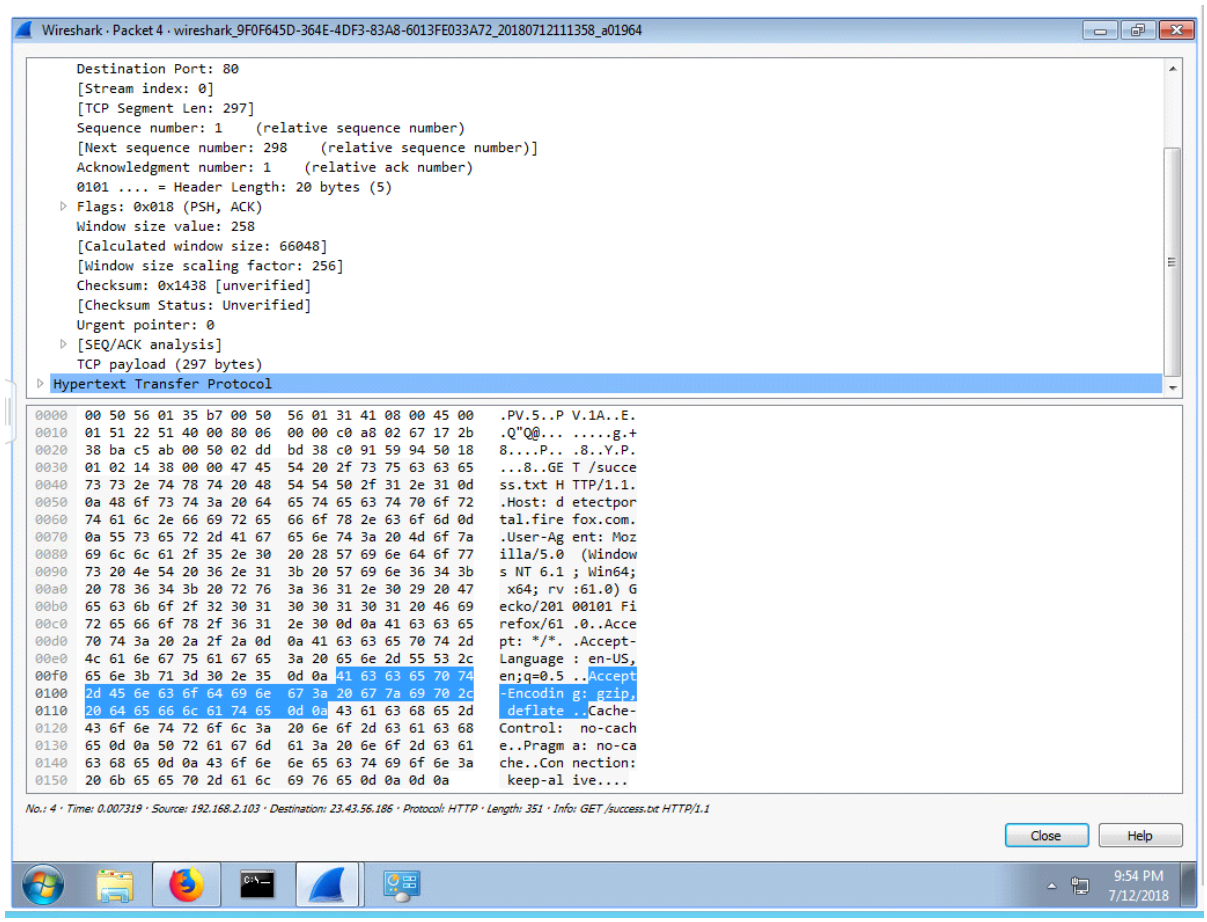
- Click on the arrow to the left of **Hypertext Transfer Protocol** entry



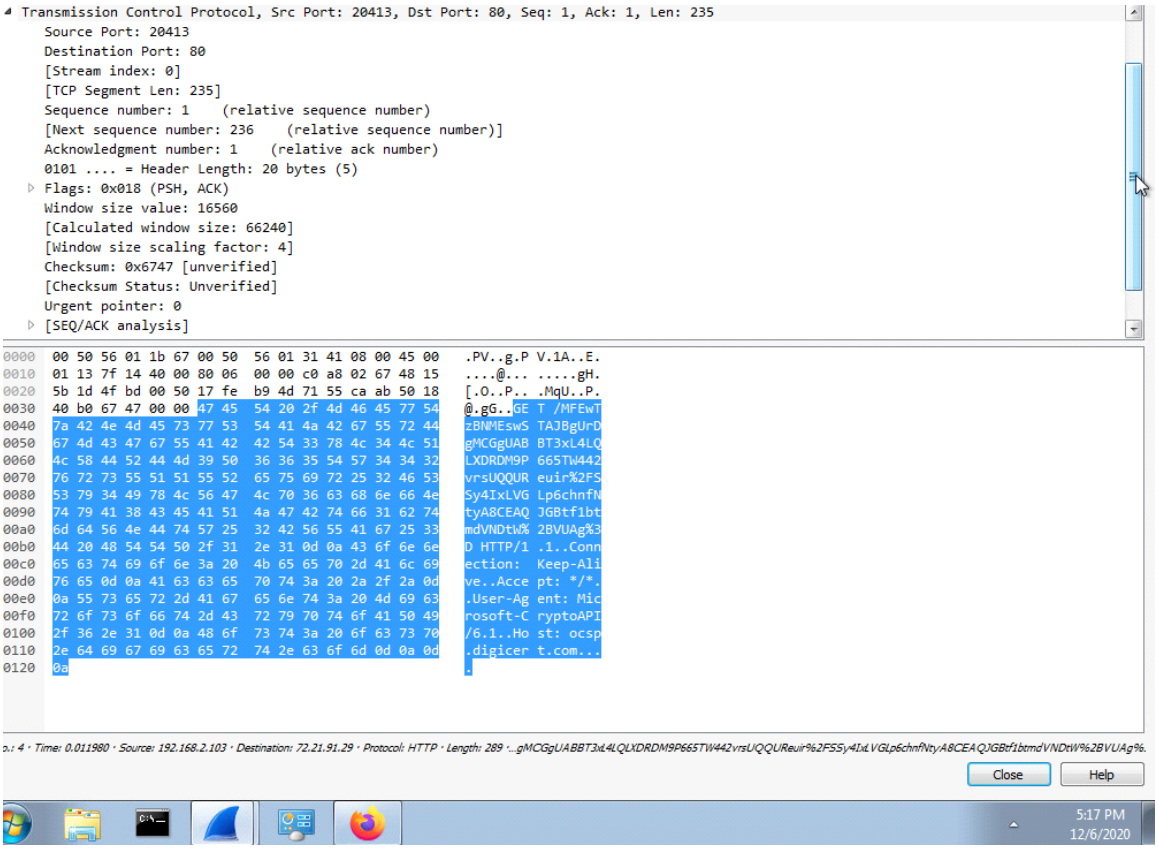
- The entry collapses
- Click on the arrow to the left of **Transmission Protocol** entry



- The **Transmission Protocol** entry expands
- Using Snipping Tool, take a screenshot that includes the date/time. It should look like



INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***15 POINTS***



- This QUESTION is worth ***10 points***. Write your answer in the box.

What is the **Time to live** value for the IP packet ? HINT: Expand the **Internet Protocol Version 4** section and look for the information

The time to live for the IP packet is 128 seconds.

- This QUESTION is worth ***10 points***. Write your answer in the box.

What are the **protocols** in the captured **frame** ? HINT: Expand the **Frame 4** information and search for the information. I am looking for a value between brackets. It has 5 fields separated by a colon

The protocols in the captured frame are ethernet based protocols ip, tcp, and http. 'eth:ethertype:ip:tcp:http'

- This QUESTION is worth ***10 points***. Write your answer in the box.

What version of HTTP is used ? HINT: Expand the **Hypertext Transfer Protocol** entry

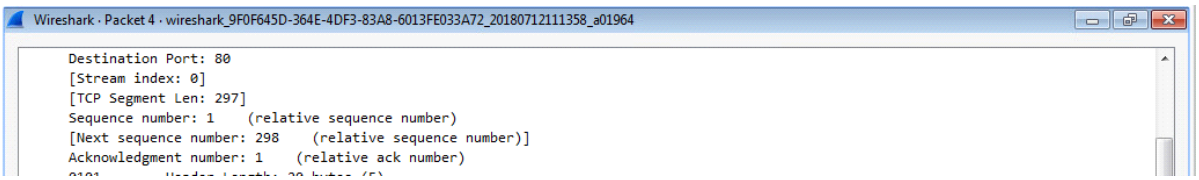
The requested version of HTTP was 1.1

- This QUESTION is worth ***10 points***. Write your answer in the box.

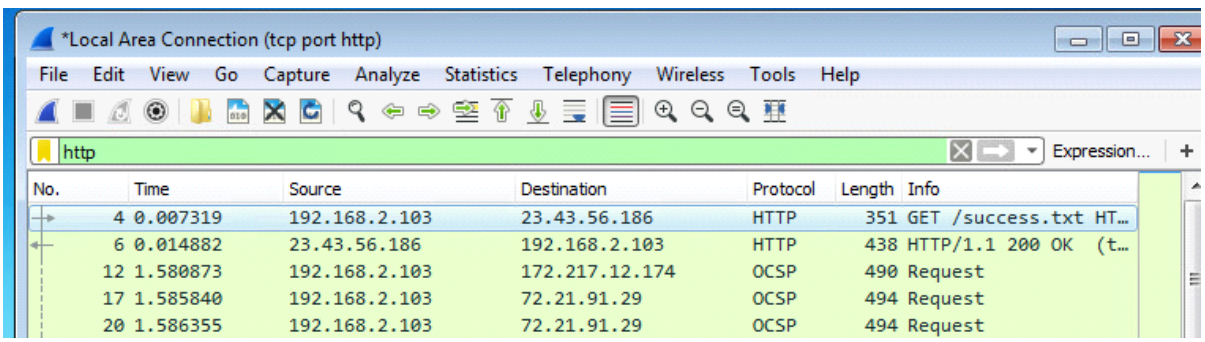
What is the size in bytes of the **TCP payload** ? HINT: Expand the **Transmission Control Protocol** entry and search for the information

The size in bytes of the TCP payload was 235 bytes.

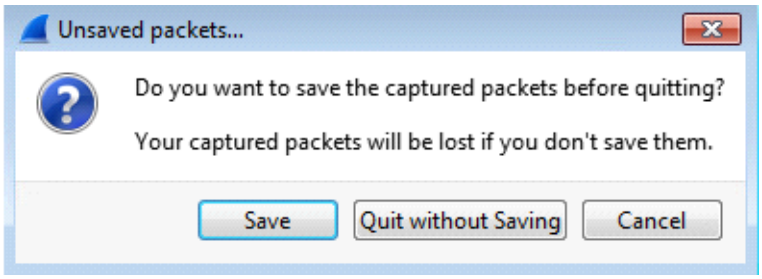
- Click on **X** at top right of the **Packet** window to close



- Click on **X** at top right of the **Local Area Connection** window to close



- A **Unsaved packets** message appears
- Click on **Quit without Saving**



- Close **Firefox**

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